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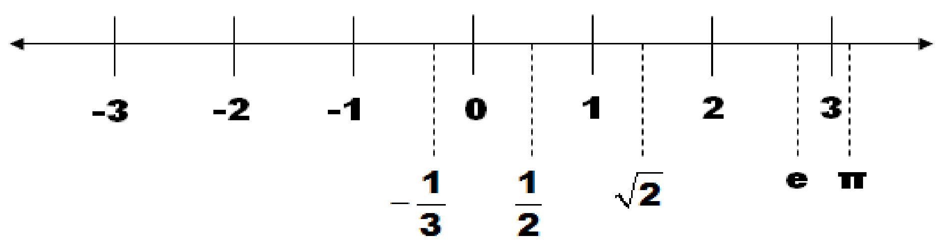
Calculus II for biological Sciences , Math 10A (First Exam Material)

1. Real Number System

A **real number** is a number that can be found on the number line. These are the numbers that we normally use and apply in real-world applications.

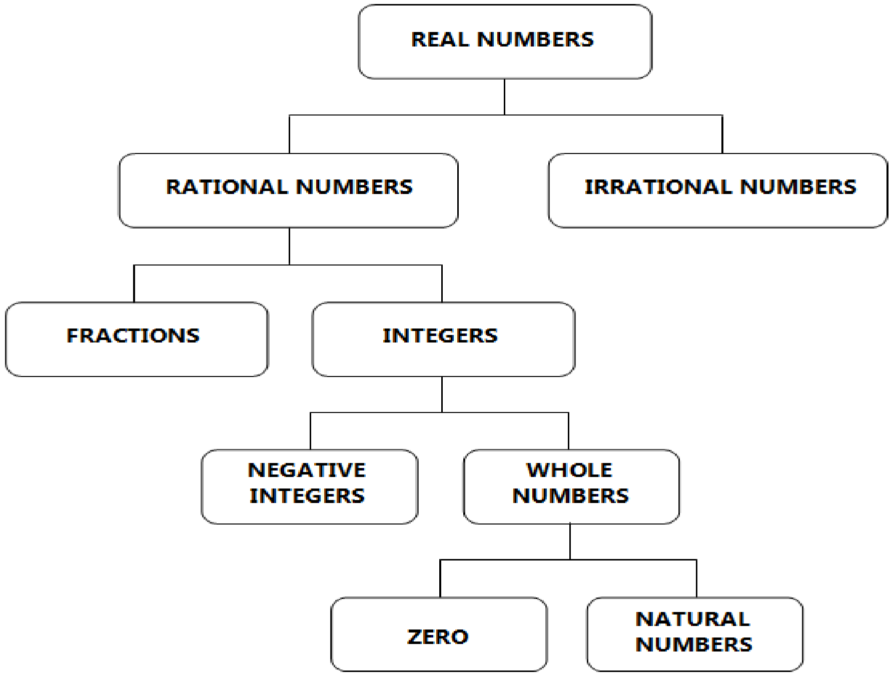


There are many types of real numbers ($\mathbb{R}$).

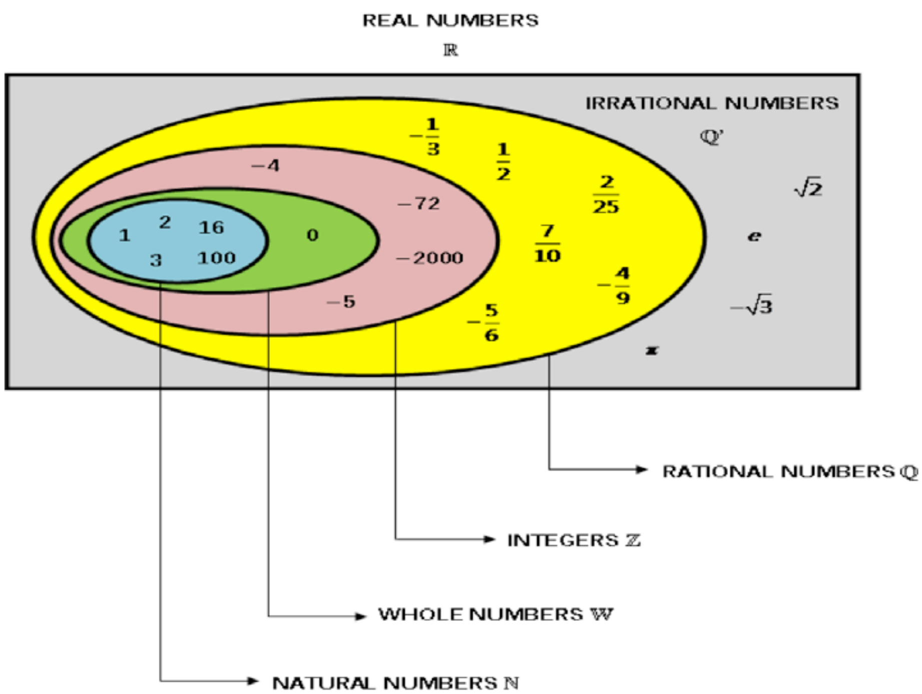
<u>Here are some of them:</u>

1. Natural numbers ($\mathbb{N}$)
2. Whole numbers
3. Integers ($\mathbb{Z}$)
4. Fractions
5. Rational numbers ($\mathbb{Q}$)
6. Irrational numbers ($\overline{\mathbb{Q}}$)

Below is a diagram of the real number system.



- Real numbers are mainly classified into rational and irrational numbers.
- Rational numbers include all integers and fractions.
- All negative integers and whole numbers make up the set of integers.
- Whole numbers comprise of all natural numbers and zero.



2. Solving linear and quadratic inequalities

A. Interval notation for the given inequality.

Inequality	Graph	Interval Notation
$a \leq x \leq b$		$[a, b]$
$a < x < b$		(a, b)
$a \leq x < b$		$[a, b)$
$a < x \leq b$		$(a, b]$
$x > a$		(a, ∞)
$x \geq a$		$[a, \infty)$
$x < b$		$(-\infty, b)$
$x \leq b$		$(-\infty, b]$

B. Properties of Inequality

For real numbers a, b and c:

1. If $a < b$, then $a + c < b + c$,
2. If $a < b$ and if $c > 0$, then $ac < bc$,
3. If $a < b$ and if $c < 0$, then $ac > bc$.
4. $a<b<c$ is equivalent to $a<b$ and $b<c$

Replacing $<$ with $>$, ≤ , or ≥ results in similar properties.

Example 1: Simplify and sketch each of the following intervals on the the real number line:

a) $(1,4]\cap[2,4]$

b) $[0,\infty)\cap(-\infty,2)$

c) $(2,3)\cup[1,5)$

d) $(2,4]\cup(4,5]$

e) $(1,3)\cap(3,5]$

Example 2: Solve each of the following inequalities

(a) $4x+5>3x+10$

(b) $(x-1)(x-2)\geq 0$

(c) $1-x\leq x+3\leq 9-2x$

(d) $2x-4<2$ or $3x-1>11$

(e) $\dfrac{2x-4}{x-3}\geq 0$

(f) $\dfrac{x}{x-1}\geq 3$

(g) $\dfrac{x^2}{x-1}<0$

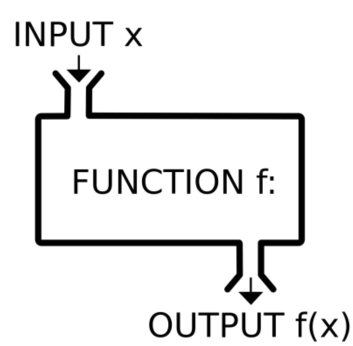
(h) $x^2+1>0$

(i) $2x+1<2x$

3. Functions

Definition: A function is a special relationship where each input has a single output.

It is often written as "y=f(x)" where x is the input value.



Example 1: Given $f(x)=5x^2+3x+2$. Find

(a) $f(1), f(1+h)$

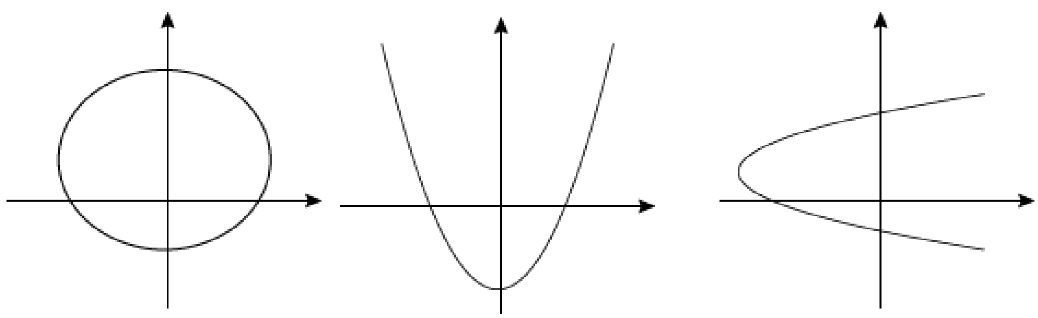
(b) $\dfrac{f(1+h)-f(1)}{h}, h\neq 0$

Vertical Line Test

On a graph, the idea of **single valued** means that no vertical line ever crosses more than one value.

If it **crosses more than once** it is still a valid curve, but is **not a function**.

Example 2: Which of the following graphs represents a function?



The Domain of a Function

"The Domain of a function is the set of all the allowable x or input values."

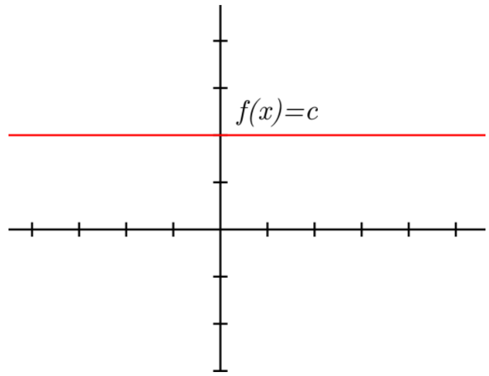
The Range of a Function

"The Range of the function is the set of resulting y or output values."

Some types of functions

A) Constant function

- Constant function is the form of $y=f(x)=c$, where c is a constant
- Domain is $\mathbb{R}=(-\infty,\infty)$
- Range is $\{c\}$
- **Graph of the constant function**



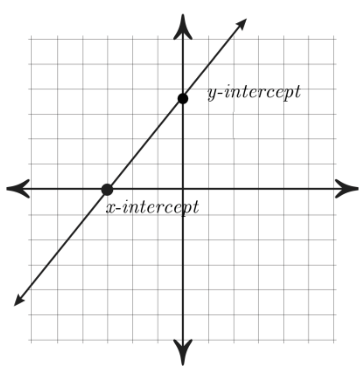
Example: Given $f(x) = 3$.

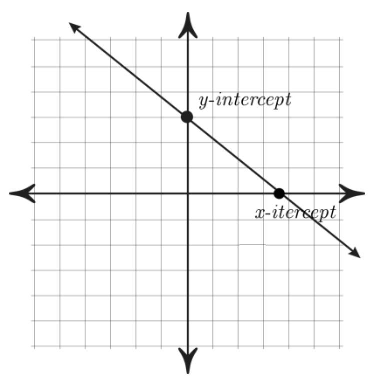
(a) Determine the domain and the range of $f(x)$

(b) Sketch of $f(x)$

B) Linear Function

- Linear function is the form of $f(x) = ax + b, a \neq 0$, where a and b are constants
- a is the slope and b is $y-$ intercept
- Domain is $\mathbb{R} = (-\infty, \infty)$
- Range is $\mathbb{R} = (-\infty, \infty)$
- **Graph of the linear function**
- **Graph is increasing if $a > 0$ and decreasing if $a < 0$**





Example: Given $f(x) = 6 - 2x$.

(a) Determine the domain and the range of $f(x)$

(b) Determine the $x-$intercept and $y-$itercept

(b) Sketch of $f(x)$

Equation of the straight line:

The slope m of the straight line passing through the points $\mathrm{P}(x_1, y_1)$ and $\mathrm{Q}(x_2, y_2)$, is

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1},\ x_1 \neq x_2$$

The equation of a straight line with slope m, passing through the point $\mathrm{P}(x_1, y_1)$, is

$$y - y_1 = m(x - x_1)$$

Example 1: Find the slope of the line joining the points $P(1,-3)$ and $Q(3,7)$.

Example 2: Find the equation of the line through the point $P(-5,3)$ whose slope is -2

Example 3: Find the equation of the straight line satisfying the condition in each of the following:

(a) passing through $P(0,0)$ with slope 3.

(b) passing through $\mathrm{P}(2,1)$ and $\mathrm{Q}(3,4)$.

(c) with slope 3 and $y-$intercept -2

Example 4: Find the slope $y-$intercept for each of the following linear relations:

(a) $3x + 5y = 15$

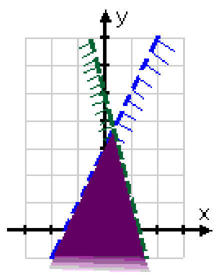
(b) $\frac{x}{5} - \frac{2y}{4} - 12 = 0$

Solving system of linear inequalities:

Example: Solve the following system

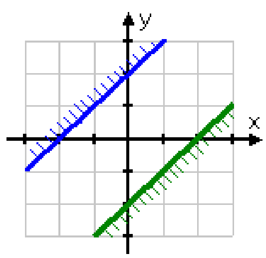
1. $2x - y > -3$ and $4x + y < 5$

   Solution:



2. $y \geq x + 2$ and $y \leq x - 2$

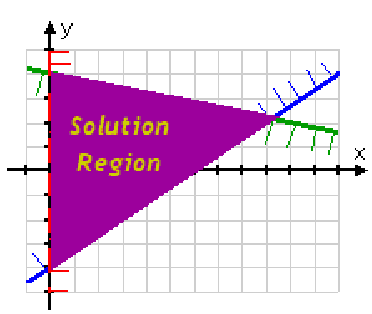
Solution



Since there is no intersection, there is **no solution.**

3. $2x - 3y \leq 12,\ x + 5y \leq 20$ and $x > 0$

   Solution



Example: Sketch the graph of the following inequalities in the $xy-$plane

(a) $x+y>1$

(b) $2x-y\le 4$

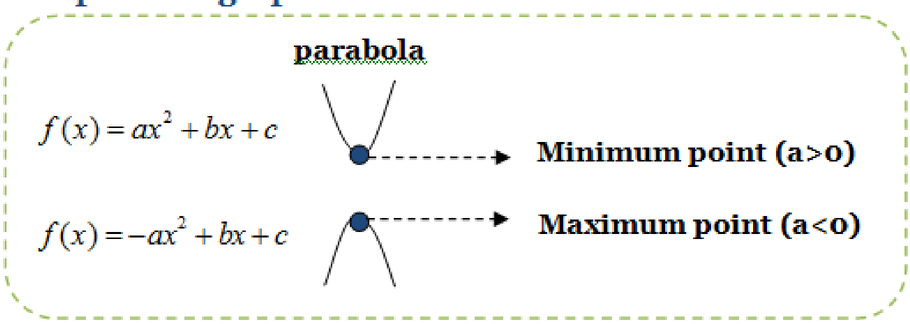
(c) $x+2y>2$ and $3x+y<3$

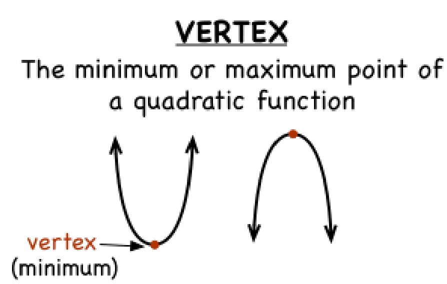
(d) $2x+y\ge 4, x+2y<4$ and $2x-3y<3$

C) Quadratic function (Parabola)

- Quadratic function is the form of $f(x)=ax^2+bx+c,\ a\ne 0$, where a,b and c are constants
- Domain is $\mathbb{R}=(-\infty,\infty)$
- Range
  - If $a>0$, then the range of $f(x)$ is $[f(-b/2a),\infty)$
  - If $a<0$, then the range of $f(x)$ is $(-\infty, f(-b/2a)]$
- **Graph of the linear function**

Shape of the graph





- **Vertex is** $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

Example: Given $f(x)=4x-x^2$.

(a) Determine the domain and the range of $f(x)$

(b) Find the vertex of $f(x)$

(c) Find the $x-$ intercepts and $y-$ intercept

(d) Sketch of $f(x)$

D) Rational Functions

Definition: A rational function $f(x)$ is a function which is the ratio of two polynomials, that is , $f(x)=\dfrac{p(x)}{q(x)}, q(x)\neq 0$ where $p(x)$ and $q(x)$ are polynomials.

For examples, $f(x)=\dfrac{x^2+2x-1}{x+6}, f(x)=\dfrac{3}{x+4}, f(x)=\dfrac{x^2+1}{x^5-3x}$

- Domain of $f(x)$ is $\mathbb{R}-\{x:q(x)=0\}$

Example: Find the domain of the following functions

(a) $f(x)=\dfrac{x-4}{x^2-2x}$ (b) $f(x)=\dfrac{2}{x^2+4}$

Special Case:

If $f(x)=\dfrac{ax+b}{cx+d}$, the domain of f is $\mathbb{R}-\{-d/c\}$ and the range of $\mathbb{R}-\{a/c\}$ such that no common factors between $ax+b$ and $cx+d$.

Example: Find the domain and the range of the following functions

(a) $f(x)=\dfrac{2x+2}{x-5}$ (b) $f(x)=\dfrac{3}{x-4}$ (c) $f(x)=\dfrac{2x-4}{x-2}$ (d) $f(x)=\dfrac{x-4}{3}$

E) Radical Function (The nth root function)

A radical is a root of a number, which can be square roots, cubic roots, and so on. A square is also called a radical. A radical function is any function that is defined in a root. This function also contains a square root, cubed roots, or any of the nth root.

General form of radical function: $f(x)=\sqrt[n]{g(x)}$

- Domain of $f(x)=\sqrt[n]{g(x)}$ is $\begin{cases} x : g(x)\geq 0, \text{if } n \text{ is even}, n\geq 2 \\ \text{domain } g(x), \text{if } n \text{ is odd}, n\geq 3 \end{cases}$
- Domain of $f(x)=\sqrt[n]{x}$ *is* $\begin{cases} [0,\infty), \text{ if } n \text{ is even}, n\geq 2 \\ (-\infty,\infty), \text{if } n \text{ is odd}, n\geq 3 \end{cases}$

Example: Find the domain of the following functions

(a) $f(x)=\sqrt{x^2+2x}$

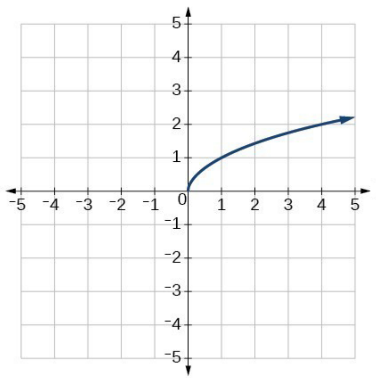
(b) $f(x)=\sqrt[3]{\dfrac{x+2}{x-3}}$

(c) $f(x)=\sqrt[4]{\dfrac{x^2}{x-2}}$

- **Domain and Range function of the form**

  $f(x) = a + \sqrt{bx + c}$ and $f(x) = a - \sqrt{bx + c}$

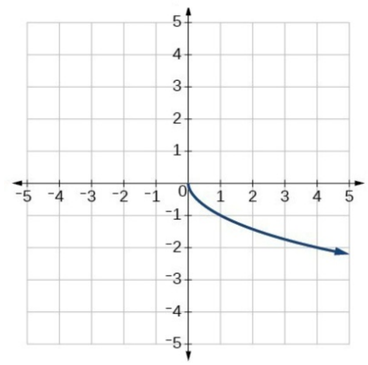
<u>**Some examples**</u>



$f(x) = \sqrt{x}$

Domain is $[0, \infty)$

Range is $[0, \infty)$



$f(x) = -\sqrt{x}$

Domain is $[0, \infty)$

Range is $(-\infty, 0]$

Example: Find the domain and range graphically for the following functions

a) $f(x) = -2 + \sqrt{2 - x}$

b) $f(x) = -\sqrt{x - 3}$

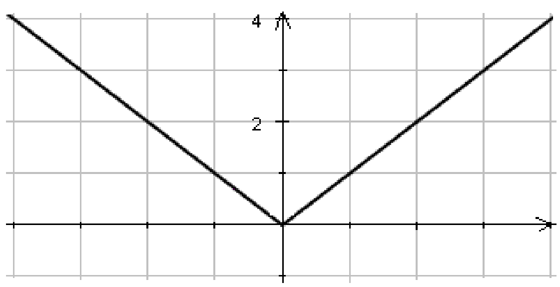
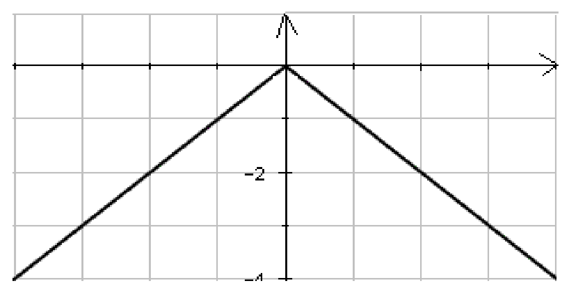
c) $f(x) = -\sqrt{1 + x}$

d) $f(x) = 4 + \sqrt{x - 3}$

F) The Absolute value function

General form of an absolute value function : $f(x) = | ax + b | + c$ or $f(x) = - | ax + b | + c$

Some examples:

$f(x) = \| x \| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ Graph is  Domain is $\mathbb{R}$ and Range is $[0, \infty)$	$f(x) = - \| x \| = \begin{cases} -x, & x \geq 0 \\ x, & x < 0 \end{cases}$ Graph is  Domain is $\mathbb{R}$ and Range is $(-\infty, 0]$

Example: Find the domain and range graphically for the following functions

a) $f(x) = -2 + | x - 4 |$

b) $f(x) = - | x + 4 |$

c) $f(x) = 2 - | 2x + 6 |$

d) $f(x) = 4 + | x |$

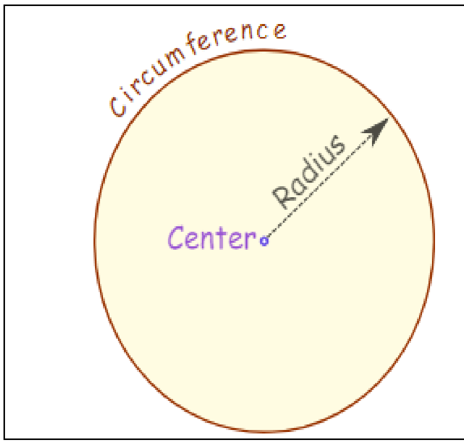
e) $f(x) = \dfrac{| x - 1 |}{x - 1}$

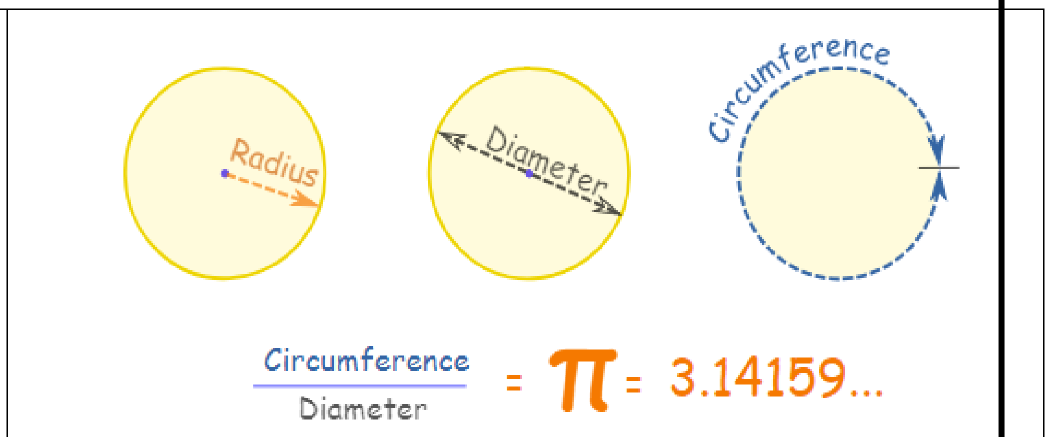
G) Upper circle function and Lower circle function

1) Circle: All points are the same distance from the center.

Radius, Diameter and Circumference

- The **Radius** is the distance from the center outwards.
- The **Diameter** goes straight across the circle, through the center.
- The **Circumference** is the distance once around the circle.





EQUATION OF A CIRCLE

The equation of a circle comes in two forms:

1) The standard form: $(x-h)^2+(y-k)^2=r^2$

2) The general form : $x^2+y^2+Dx+Ey+F=0$ where D, E and F are constants.

Example: Find an equation of the circle with radius 3 and center (2, -5).

Example: Find the equation of a circle with center at the origin and passing through

Example: Determine the equation for a circle in standard form with a radius of 3, and centered at the point (-3,4).

Example: Find the center and radius of the circle having the following equation:

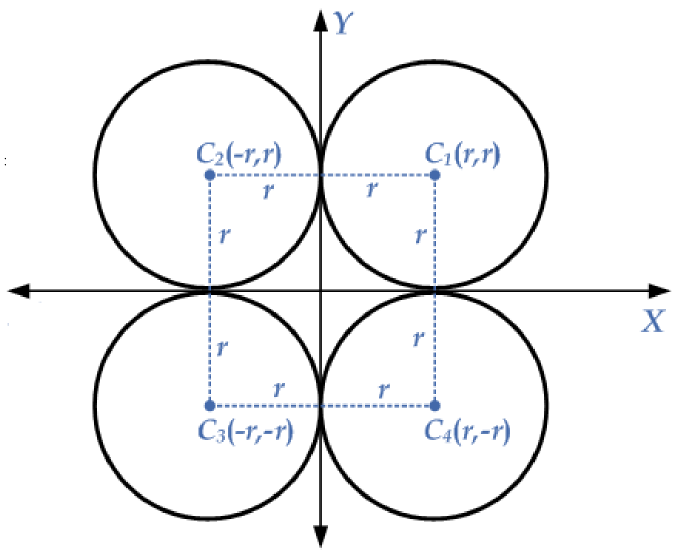
$4x^2+4y^2-16x-24y+51=0$

Example: Find the center and radius of the circle having the following equation:

$x^2 + y^2 - 4x + 6y - 11 = 0$

<u>Special Cases:</u>

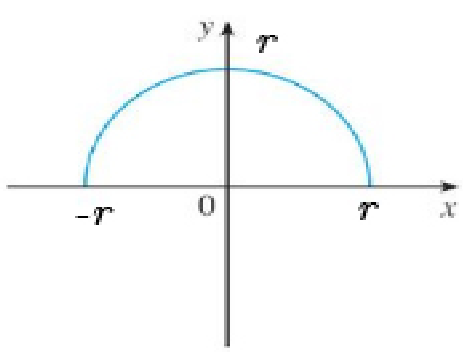
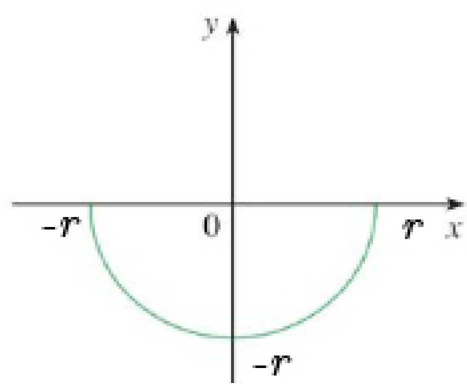
$(x+r)^2 + (y-r)^2$ $\quad\quad$ $+(y-r)^2 = r^2$



$(x+r)^2 + (y+r)^2$ $\quad\quad$ $+(y+r)^2 = r^2$

Example: Find the equation of circle that lies in the first quadrant has a radius of 2 units and touches the coordinate axis?

2) **<u>Upper circle function and Lower circle function</u>**

 $f(x) = \sqrt{r^2 - x^2}$ Domain is $[-r, r]$ Range is $[0, r]$	 $f(x) = -\sqrt{r^2 - x^2}$ Domain is $[-r, r]$ Range is $[-r, 0]$

Example: Sketch and find the domain and the range of the following functions

(a) $f(x) = \sqrt{9 - x^2}$

(b) $f(x) = -\sqrt{16 - x^2}$

(c) $f(x) = 5 + \sqrt{4 - x^2}$

(d) $f(x) = -3 - \sqrt{1 - x^2}$

<u>H) Composition function</u>

Given two functions f and g, the composition of the f and g denoted by $f \circ g$ is defined $(f \circ g)(x) = f(g(x))$

- The domain of $f \circ g$ is given by $\{x : x \in \text{Domain of } g \text{ and } g(x) \in \text{Domain of } f\}$

Example: Given $f(x) = x^2$ and $f(x) = \sqrt{x - 1}$. Find

(a) $(f \circ g)(5)$ (b) $(g \circ f)(3)$

Example: Let $f(x) = \dfrac{1}{x - 1}$ and $g(x) = \dfrac{3}{x - 2}$. Find

(a) $(f \circ g)(x)$ (b) domain of $(f \circ g)(x)$

Example: Find $g(x)$ if

(a) $f(x) = x^2$ and $(f \circ g)(x) = (1 + x)^2$

(b) $f(x) = \dfrac{x + 1}{x + 2}$ and $(g \circ f)(x) = \dfrac{3}{x - 2}$

Example: Find $g(0.5)$ if $f(x) = \dfrac{x + 1}{x + 2}$ and $(g \circ f)(x) = \dfrac{3}{x - 2}$

I) Arithmetic Operations on functions

1. **Sum:** $(f+g)(x) = f(x) + g(x)$ **,** domain: the intersection of the domains of f and g
2. **Difference:** $(f-g)(x) = f(x) - g(x)$, domain: the intersection of the domains of f and g
3. **Product:** $(f \times g)(x) = f(x) \times g(x)$, domain: the intersection of the domains of f and g
4. **Quotient:** $(f \,/\, g)(x) = f(x) \,/\, g(x)$, domain: the intersection of the domains of f and g with the points where $g(x) = 0$ excluded.

Example: Find the domain of the following

(a) $f(x) = \sqrt{x-1} + \sqrt{9-x^2}$

(b) $f(x) = \dfrac{\sqrt{x^2-16}}{x^2-25}$

H) Limits at Infinity

Fact 1

1. If r is a positive rational number and c is any real number then, $\lim\limits_{x\to\infty} \dfrac{c}{x^r} = 0$

2. If r is a positive rational number and c is any real number and x^r is defined for $x < 0$ then, $\lim\limits_{x\to-\infty} \dfrac{c}{x^r} = 0$

Fact 2

If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ is a polynomial of degree n (i.e. $a_n \neq 0$) then,

$\lim\limits_{x\to\infty} p(x) = \lim\limits_{x\to\infty} a_n x^n$, $\lim\limits_{x\to-\infty} p(x) = \lim\limits_{x\to-\infty} a_n x^n$